

Spread Spectrum (SS) and CDMA Systems

11.1 Introduction

We introduced spread spectrum techniques in Chapter 6. In this chapter, we present details of direct-sequence spread spectrum (DSSS) and frequency-hop spread spectrum (FHSS) systems [1,2,4]. We show how signal spreading and despreading is achieved with binary phase shift keying (BPSK) and quadrature phase shift keying (QPSK) modulation in the DSSS [11,12]. We then address multipath issues in wireless communications and show how code division multiple access (CDMA) takes advantage of multipath in improving system performance with a Rake receiver [6,13]. We conclude the chapter by presenting a summary of the challenges in implementing a CDMA system and providing some highlights of the Telecommunication Industries Association (TIA) IS-95 CDMA system. Those who are not familiar with spreading codes should refer to Appendix D.

11.2 Concept of Spread Spectrum

In a wideband spread-spectrum (SS) system, the transmitted signal is spread over a frequency band that is much larger, in fact, than the maximum bandwidth required to transmit the information bearing (baseband) signal [3]. An SS system takes a baseband signal with a bandwidth of only a few kilohertz (kHz), and spreads it over a band that may be many megahertz (MHz) wide. In SS systems, an advantage in signal-to-noise ratio (SNR) is achieved by the modulation and demodulation process.

The SS signal is generated from a data-modulated carrier. The data-modulated carrier is modulated a second time by using a wideband spreading signal. An SS signal has advantages in the areas of security, resistance to narrowband jamming, resistance to multipath fading, and supporting multiple-access techniques.

The spreading modulation may be phase modulation or a rapid change of the carrier frequency, or it may be a combination of these two schemes. When spectrum spreading is performed by phase modulation, we call the resultant signal a *direct-sequence spread spectrum* (DSSS) signal (see Figure 11.1) [15]. When spectrum spreading is achieved by a rapid change of the carrier frequency, we refer to the resultant signal as a *frequency-hop spread spectrum* (FHSS) signal [14]. When both direct-sequence and frequency-hop techniques are employed, the

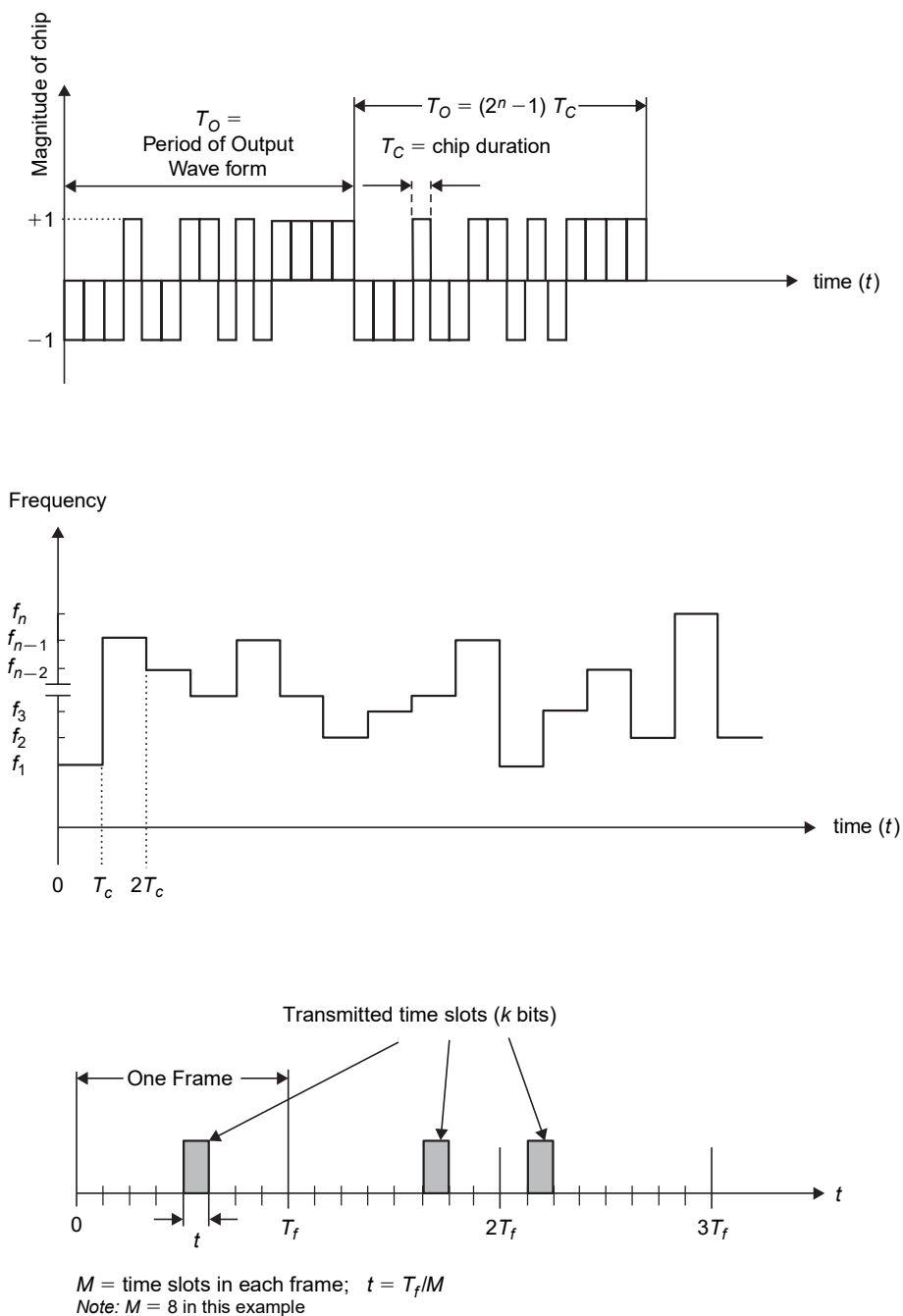


Figure 11.1 Spread spectrum techniques.

resultant signal is called a hybrid DS-FH SS signal. Another way to also generate an SS signal is the *time-hop spread spectrum* (THSS) signal. In this case, the transmission time is divided into intervals called frames. Each frame is further divided into time slots. During each frame, one and only one time slot is modulated with a message (details of THSS are not given in this chapter). The DSSS is the averaging technique to reduce interference whereas FHSS and THSS are the avoiding techniques to minimize interference.

The spreading signal is selected to have properties to facilitate demodulation of the transmitted signal by the intended receiver, and to make demodulation by an unintended receiver as difficult as possible. These same properties also make it possible for the intended receiver to differentiate between the communication signal and jamming. If the bandwidth of the spreading signal is large relative to the data bandwidth, the spread-spectrum transmission bandwidth is dominated by the spreading signal and is independent of the data signal bandwidth.

Consider the channel capacity as given by the Shannon equation [10]:

$$C = B_w \log_2(1 + S/N) \quad (11.1)$$

where:

B_w = channel bandwidth in Hertz (Hz)

C = channel capacity in bits per second (bps)

S = signal power

N = noise power

Equation 11.1 provides the relationship between the theoretical ability of a communication channel to transmit information without errors for a given signal-to-noise ratio and a given bandwidth of the channel. The channel capacity can be increased by increasing the channel bandwidth, the transmitted power, or a combination of both.

Shannon modeled the channel at baseband. However, Equation 11.1 is applicable to a radio frequency (RF) channel by assuming that the intermediate frequency (IF) filter has an ideal (flat) bandpass response with a bandwidth that is at least $2B_w$. This bound assumes that channel noise is additive white Gaussian noise (AWGN). AWGN is often used in the modeling of an RF channel. This assumption is justified since the total noise is generated by random electron fluctuations. The central limit theorem provides us with the assumption that the output of an IF filter has a Gaussian distribution and is frequency independent. For most communications systems that are limited by thermal noise, this assumption is valid. For interference-limited systems, this assumption is not valid and the results may be different. The Shannon equation does not provide a method to achieve the bound. Approaching the bound requires complex channel coding and modulation schemes. In many cases, achieving an implementation that provides performance near this bound is impractical due to the resulting complexity. Spread spectrum

systems can be engineered to operate at much lower SNRs since the channel bandwidth can be traded for the SNR to achieve good performance at a very low SNR.

We rewrite Equation 11.1 as:

$$\frac{C}{B_w} = 1.44 \log_e(1 + S/N) \quad (11.2)$$

Since

$$\log_e(1 + S/N) = \frac{S}{N} - \frac{1}{2} \left(\frac{S}{N} \right)^2 + \frac{1}{3} \left(\frac{S}{N} \right)^3 - \frac{1}{4} \left(\frac{S}{N} \right)^4 + \dots \quad (11.3)$$

Assuming that the S/N ratio is small (e.g., $S/N \leq 0.1$), we can neglect the higher-order terms and rewrite Equation 11.2 as:

$$B_w \approx \frac{C}{1.44} \times \frac{1}{(S/N)} \quad (11.4)$$

For any given S/N ratio we can have a low information error rate by increasing the bandwidth used to transmit the information. As an example, if we wish a system to operate on an RF link in which the data information rate is 20 kilobits per second (kbps) and the S/N ratio is 0.01, we should use a bandwidth of

$$B_w = \frac{20 \times 10^3}{1.44 \times 0.01} = 1.38 \times 10^6 \text{ Hz or } 1.38 \text{ MHz} \quad (11.5)$$

Information can be modulated into the SS signal by several methods. The most common method is to multiply the information with the spread-spectrum code (refer to Appendix D) before it is used to modulate the carrier frequency (see Figure 11.2). This technique applies to any SS system that uses a spreading code to determine RF bandwidth. If the signal that is being sent is analog (voice, for example), the signal must be digitized before being modulated by the spreading code.

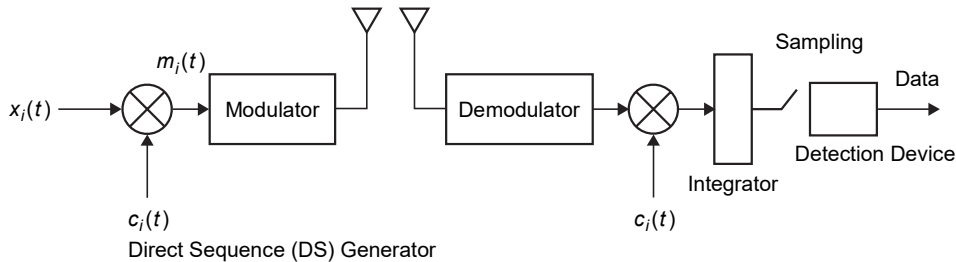


Figure 11.2 Direct sequence spread spectrum system.

11.3 System Processing Gain

In the DSSS system the baseband signal is spread over a large bandwidth by a spreading code. One of the major advantages of a DSSS system is the robustness to interference [8,9]. The system processing gain, G_p , quantifies the degree of interference rejection. The system processing gain is the ratio of spreading rate, R_c , to the information rate, R_b , and is given as:

$$G_p = R_c/R_b \quad (11.6)$$

where:

R_b = information bit rate

R_c = spreading rate

Typical processing gains of a DSSS system lie between 10 and 60 dB. With a DSSS system, the total noise level is determined both by thermal noise and by interference. For a given user, the interference is processed as noise. The input and output S/N ratios are related as:

$$(S/N)_o = G_p(S/N)_i \quad (11.7)$$

We express input $(S/N)_i$ ratio as:

$$\left(\frac{S}{N}\right)_i = \left(\frac{E_b \times R_b}{N_0 \times B_w}\right)_i = \left(\frac{E_b}{N_0}\right)_i \times \frac{1}{G_p} \quad (11.8)$$

where:

E_b = bit energy

N_0 = noise power spectral density

Using Equation 11.8 we rewrite Equation 11.7 as:

$$\left(\frac{S}{N}\right)_o = G_p \cdot \left(\frac{S}{N}\right)_i = \left(\frac{E_b}{N_0}\right)_i \quad (11.9)$$

but

$$\left(\frac{S}{N}\right)_o = \left(\frac{E_b \times R_b}{N_0 \times B_w}\right)_o = \left(\frac{E_b}{N_0}\right)_o \cdot \frac{1}{G_p} \quad (11.10)$$

Therefore using Equation 11.9 and Equation 11.10 we get,

$$\left(\frac{E_b}{N_0}\right)_o = G_p \cdot \left(\frac{E_b}{N_0}\right)_i \quad (11.11)$$

Example 11.1

A DSSS system has a 1.2288 megachips per second (Mcps) code clock rate and a 9.6 kbps information rate. Calculate the processing gain. How much improvement in information rate is achieved if the code generation rate is changed to 5 Mcps and the processing gain to 256?

Solution

$$\frac{1.2288 \times 10^6}{9.6 \times 10^3} = 128 = 10 \log 128 = 21 \text{ dB}$$

$$R_b = \frac{R_c}{G_p}$$

$$R_b = \frac{5 \times 10^6}{256} = 19.53 \text{ kbps}$$

Improvement in information rate = $19.53 - 9.6 = 9.93 \text{ kbps}$

In a DSSS system the information signal, $x_i(t)$, is multiplied by a wideband code signal, $c_i(t)$, which is the output signal of the direct sequence (DS) generator (see Figure 11.2) [7,9]. The signal $x_i(t) \times c_i(t) \equiv m_i(t)$ is modulated and transmitted. This signal occupies a bandwidth far in excess of the minimum bandwidth required to transmit the information signal $x_i(t)$. We can observe from Figure 11.3 that the combined signal waveform has more high frequency changes in the data information since $1/T_c \gg 1/T_b$, where T_b is the bit interval of the information stream and T_c is the bit interval of the DSSS stream. T_c is called a *chip interval*. In a DSSS system, the pulse shape is a sequence of short rectangular pulses called *chips*. The chips are a pseudo-random sequence of 1's and -1's known at the receiver and often having a repetition period equal to the symbol period.

$$G_p = \frac{T_b}{T_c}$$

Since $R_b = 1/T_b$ and $B_w = 1/T_c$,

$$G_p = \frac{T_b}{T_c} = \frac{B_w}{R_b}$$

When $x_i(t)$ and $c_i(t)$ have the same rate, the product $m_i(t)$ contains all the information of $x_i(t)$ and has the same rate as $c_i(t)$. The spectrum of the signal remains unchanged, and the incoming bit stream is said to be encrypted or *scrambled*. However, when $c_i(t)$ has a higher rate than $x_i(t)$, $m_i(t)$ contains all the information of $x_i(t)$, has a higher bit rate compared to $x_i(t)$, and is said to have had its spectrum spread (refer to Figure 11.3).

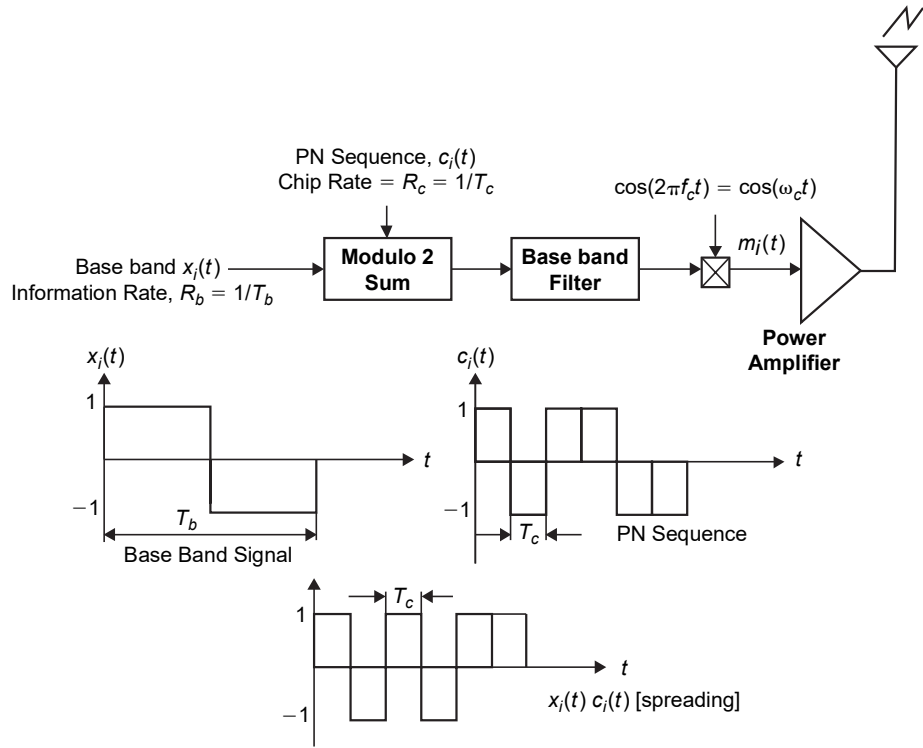


Figure 11.3 Direct sequence (DS) spreading.

Let us consider the downlink CDMA operation, in which the base station generates a data stream for mobile 1, 2, and 3 ... $x_i(t)$, and multiplies the data stream by an appropriate DS code, $c_i(t)$. Next, we add the coded data streams (see Figures 11.4, 11.5, and 11.6):

$$z(t) = \sum_{j=1}^3 m_j \quad (11.12)$$

We modulate the resultant baseband spread spectrum signal, Equation 11.12, by a carrier (frequency $\omega_c = 2\pi f_c$) to obtain

$$z_i(t) = A_i \left\{ \sum_{j=1}^3 c_j(t) x_j(t) \right\} \cos \omega_c t \quad (11.13)$$

The spread signal at the receiver for mobile, i , is $z_i(t) + \text{noise}$, where $z_i(t)$ is transmitted over the allocated bandwidth.

$$z_i(t) = A_i \left[\sum_{j=1}^3 c_j(t - \tau_j) x_j(t - \tau_j) \right] \cos(\omega_c t + \phi_i) \quad (11.14)$$

Equation 11.14 contains the desired user signal and other users' signals. After multiplication by a coherent carrier phase ϕ_i estimated by the receiver, a locally generated DS sequence that is the exact replica of the desired code transmitted multiplies the incoming signal—despreading. We assume that this DS sequence is in perfect synchronization (receiver estimates delay τ_j) with the transmitted signal so that:

$$c_j(t - \tau_j) c_j(t - \tau_j) = 1 \quad (11.15)$$

The multiplier output yields the desired data signal $x_i(t - \tau_j)$ plus interfering terms due to other users. Ideally, the integrator, an *integrate-and-dump* over T_b , should produce a cross-correlation between the desired signal and interferers that is 0. Hence, the output for mobile i is the transmitted data stream, $x_i(t - \tau_j)$.

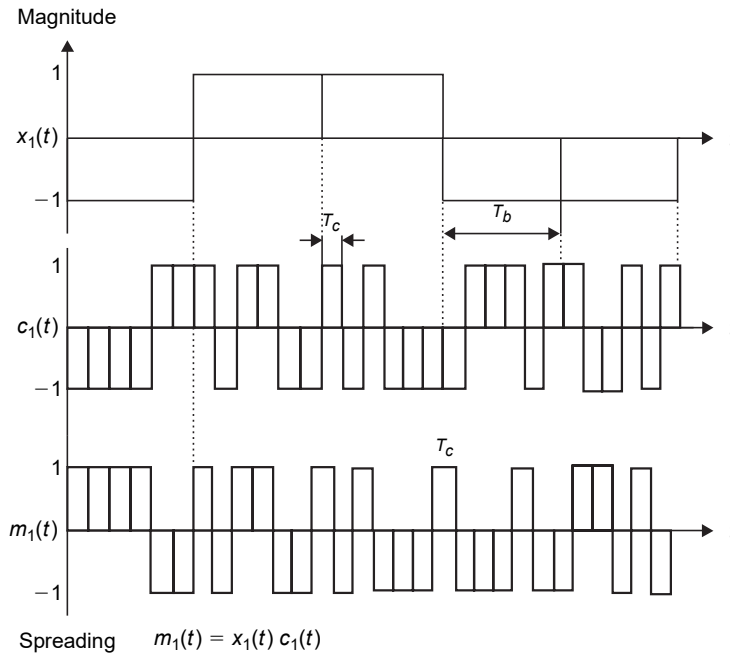


Figure 11.4 Downlink DSSS operation—signals for mobile 1, 2, and 3.

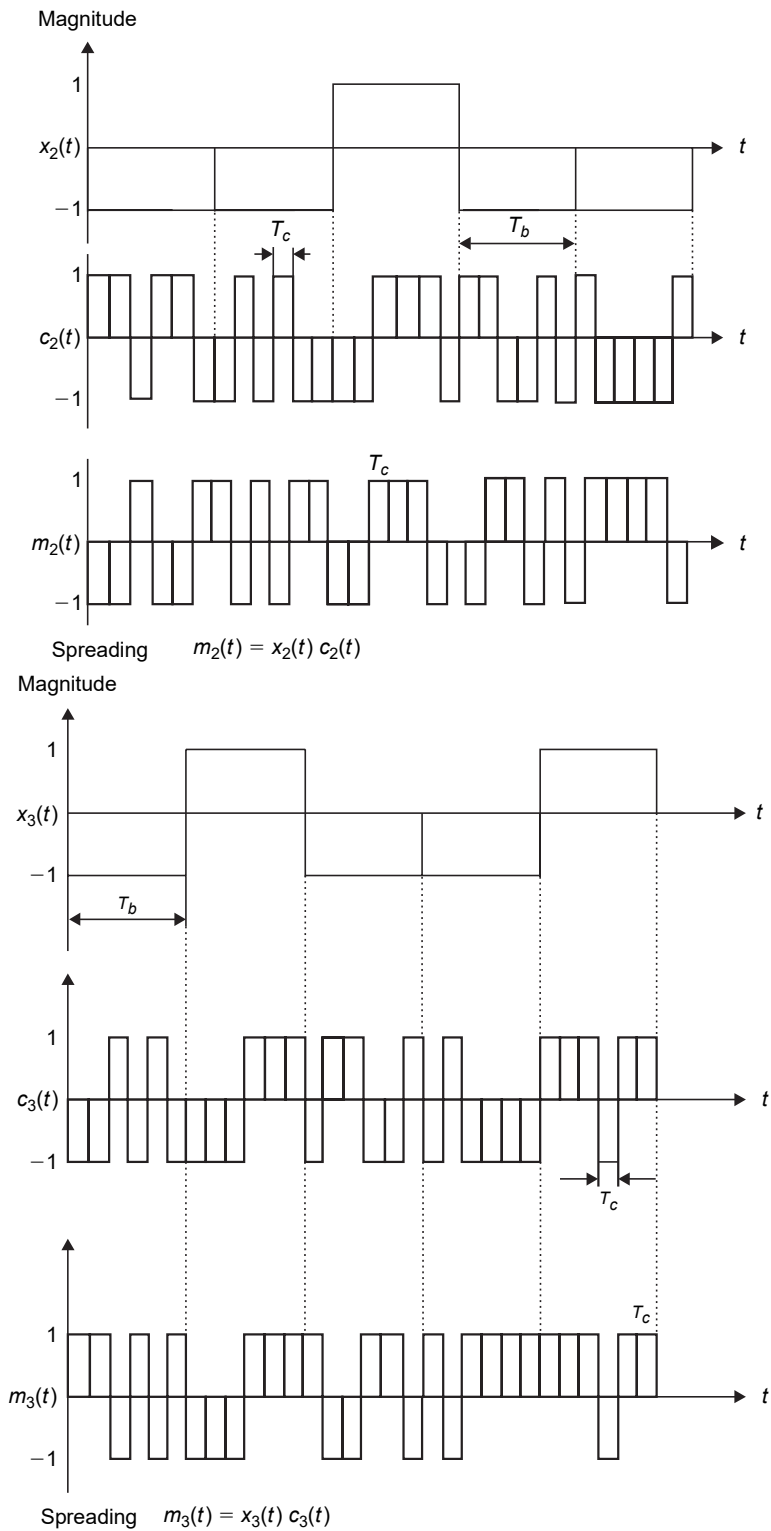


Figure 11.4 (Continued).

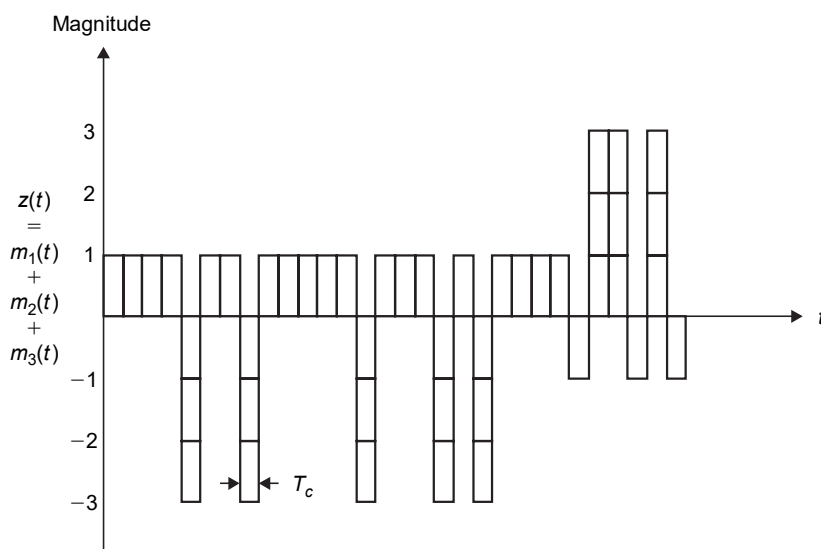


Figure 11.5 Resultant demodulated signal at a mobile.

Example 11.2

We consider signals transmitted from a base station to mobile 1, 2, and 3. The data streams $x_1(t)$, $x_2(t)$, and $x_3(t)$ are multiplied with codes $c_1(t)$, $c_2(t)$, and $c_3(t)$, respectively (see Figure 11.4). The resultant demodulated signal $z(t)$, sum of $m_1(t)$, $m_2(t)$, and $m_3(t)$ is given in Figure 11.5. Show that the transmitted signals to mobiles 1, 2, and 3 are recovered at the mobile receivers by despreading the resultant signal $z(t)$. Neglect propagation delay.

Solution

From Tables 11.1, 11.2, and 11.3, we observe that the transmitted signals for mobile 1, 2, and 3 are recovered by despreading at the receivers.

Table 11.1 $z(t)c_1(t)$ [see Figure 11.6(a)].

	T_b	$2T_b$	$3T_b$	$4T_b$	$5T_b$
Value of integration at end of bit period	-6	4	6	-2	-12
Bit value (if value of integration is <0 , then bit value equals 1)	1	0	0	1	1

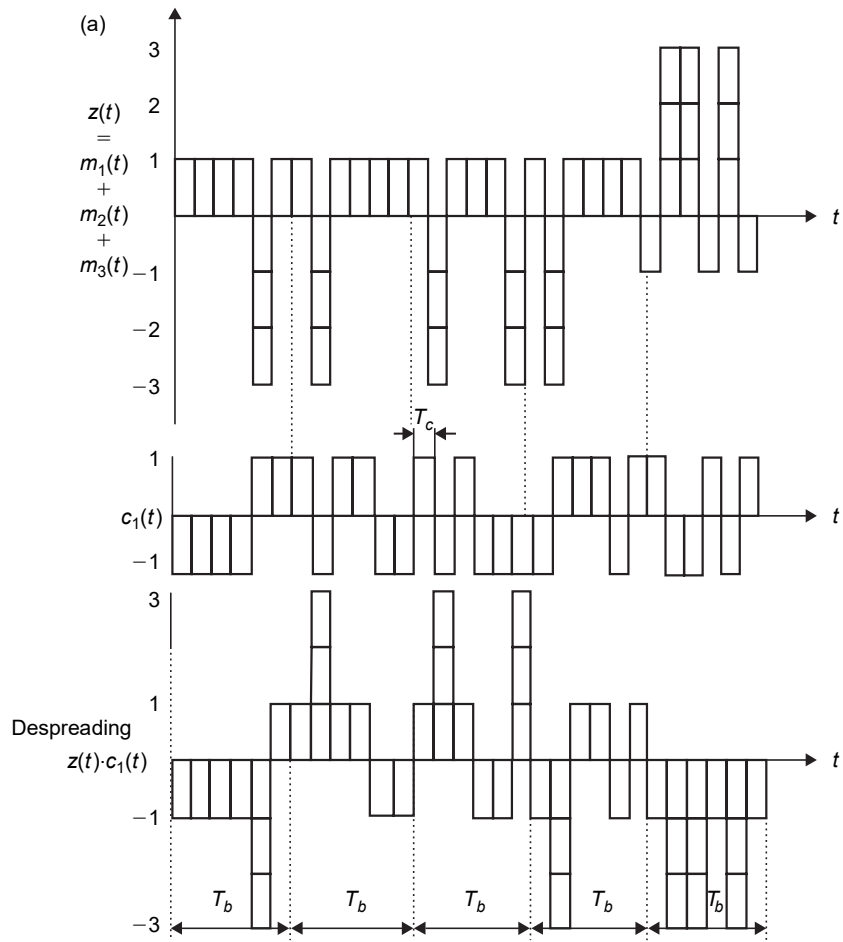


Figure 11.6(a) Despreading of resultant demodulated signal at mobile receivers.

Table 11.2 $z(t)c_2(t)$ [see Figure 11.6(b)].

	T_b	$2T_b$	$3T_b$	$4T_b$	$5T_b$
Value of integration at end of bit period	-2	-6	8	-4	-8
Bit value	1	1	0	1	1

Table 11.3 $z(t)c_3(t)$ [see Figure 11.6(c)].

	T_b	$2T_b$	$3T_b$	$4T_b$	$5T_b$
Value of integration at end of bit period	-6	4	-8	-8	8
Bit value	1	0	1	1	0

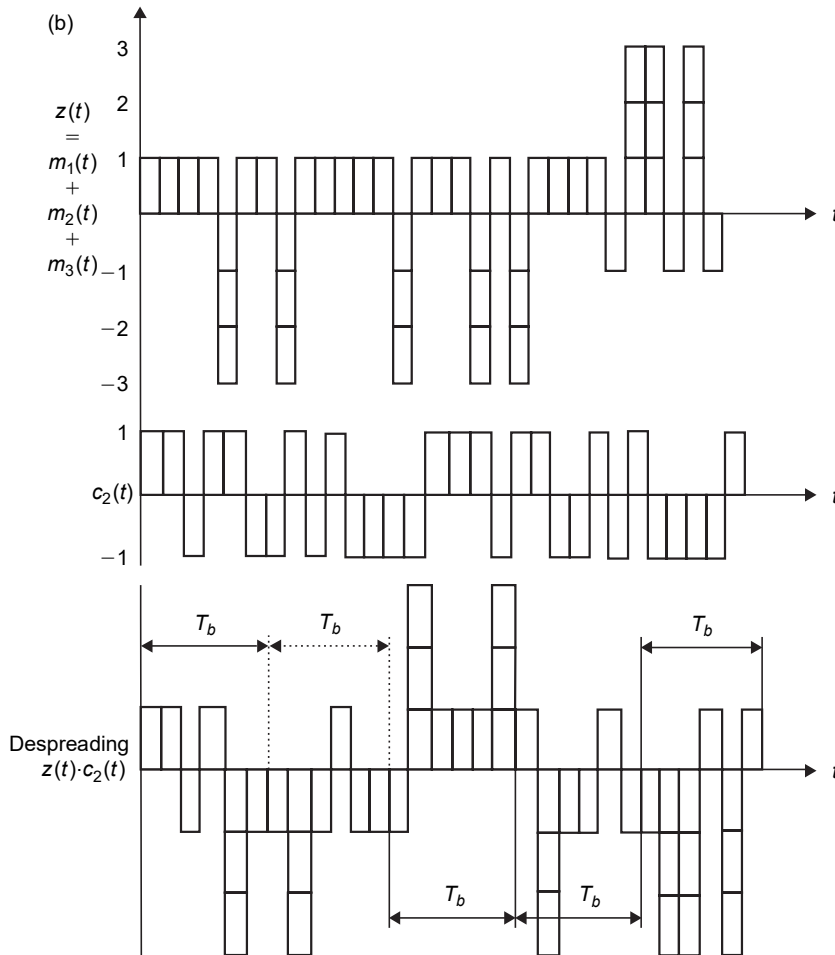


Figure 11.6(b) (Continued).

11.4 Requirements of Direct-Sequence Spread Spectrum

In the DSSS system, the entire bandwidth of the RF carrier is made available to each user. The DSSS system satisfies the following requirements [16]:

- The spreading signal has a bandwidth much larger than the minimum bandwidth required to transmit the desired information which, for a digital system, is baseband data.
- The spreading of the information is performed by using a spreading signal, called the *code signal* (see Appendix D). The code signal is independent of the data and is of a much higher chip rate than the data signal.

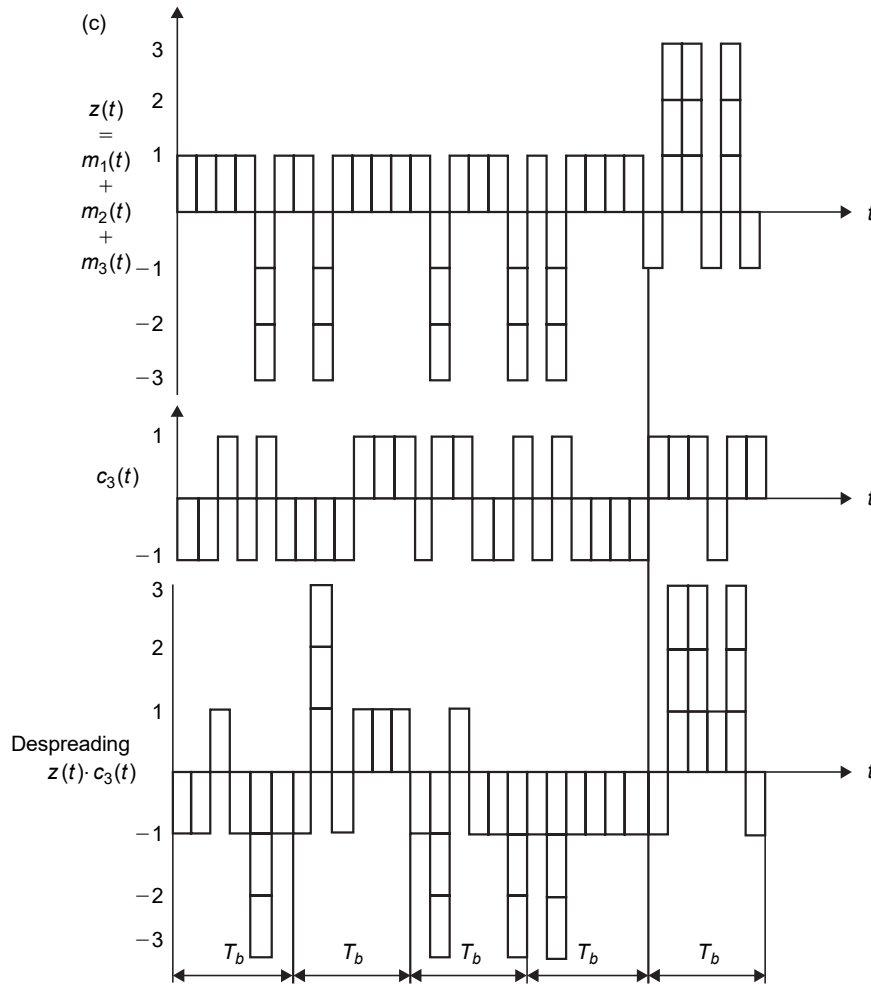


Figure 11.6(c) (Continued).

- At the intended receiver, despreading is accomplished by cross-correlation of the received spread signal with a synchronized replica of the same code signal used to spread the data.

11.5 Frequency-Hopping Spread Spectrum Systems

In FHSS systems, the binary pseudo-random noise (PN) code generator drives the frequency synthesizer to hop to one of the many available frequencies chosen by the PN sequence generator. When the hopping rate is higher than the symbol rate,

we have a *fast frequency-hop* system. If the hopping rate is lower than the symbol rate, i.e., there are several symbols transmitted per frequency hop, we have a *slow frequency-hop* system.

In an FHSS system b bits are used as an integer index for selecting the hop frequency. The term *chip* is also used in FHSS, but its meaning is different from the DSSS meaning of the word. It depends on whether the system is a slow frequency hop (FH) or a fast FH. In FH/ M -ary frequency shift keying (MFSK) a system chip is the *tone of shortest duration*. The chip rate, R_c , is the maximum of R_s and R_b , where R_s is the symbol rate and R_b is the hop rate. Thus, for slow FHSS systems, the chip rate is equal to the symbol rate (i.e., $R_c = R_s$), whereas for fast FHSS systems, the chip rate is equal to the hop rate (i.e., $R_c = R_b$).

In a slow FHSS system with MFSK, the selected M frequencies must be an integer number of symbol rates apart. The spacing is required to maintain orthogonality between the frequencies and to allow reliable noncoherent detection. This implies that the minimum bandwidth of an MFSK signal should be about MR_s .

The modulation scheme commonly used with a fast FHSS system is MFSK, where $b = \log_2 M$ bits are used to determine which one of M frequencies should be used. The position of the M -ary signal set is shifted pseudo-randomly by the frequency synthesizer over a hopping bandwidth B_w . The frequency synthesizer produces a transmission tone based on the simultaneous instructions of the PN code and the data. At each frequency hop-time a PN generator feeds the frequency synthesizer a frequency word (a sequence of b chips) which decides one

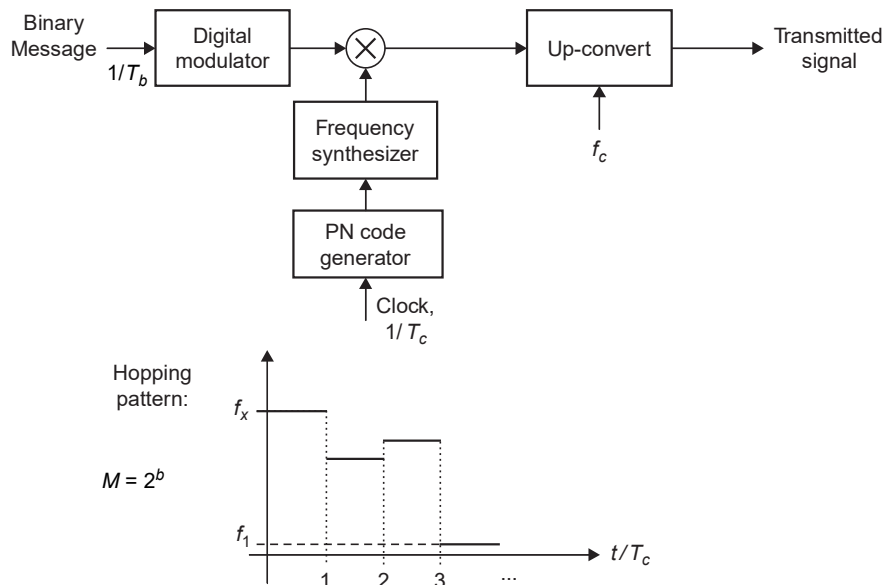
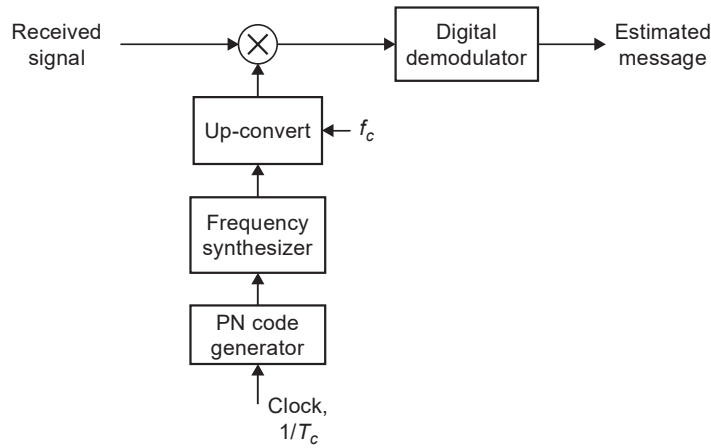


Figure 11.7 FHSS system.

**Figure 11.7 (Continued).**

of 2^b symbol-set positions (see Figure 11.7). The frequency hopping bandwidth, B_w , and the minimum frequency spacing between consecutive hop positions, Δf , determine the minimum number of chips necessary in the frequency word. The processing gain, G_p , of an FHSS system is given as:

$$G_p = \frac{\text{Hopping Bandwidth}}{\text{Minimum Frequency Spacing}} = \frac{M \cdot \Delta f}{\Delta f} = M \quad (11.16)$$

In the fast frequency-hop systems, there are L frequency hops during a symbol interval (T_s) (i.e., $T_s = LT_c$ or $R_c = LR_s$)

$$\therefore \text{Hopping Bandwidth} = (KM\Delta f)L \quad (11.17)$$

where:

K = factor for frequency multiplication

$M = 2^b$ is the number of frequencies produced by frequency synthesizer

b = bit in a symbol

L = frequency hops per symbol

$$\therefore G_p = \frac{KM\Delta f L}{\Delta f} = MKL \quad (11.18)$$

The processing gain of a fast frequency hop system is dependent upon the number of frequencies used (M), the number of hops per symbol (L), and the frequency multiplication factor (K).

Example 11.3

In an FHSS system, a hopping bandwidth of 100 MHz and a frequency spacing of 10 kHz is used. What is the minimum number of PN chips that are required for each frequency symbol?

Solution

$$\text{Number of frequency tones in hopping bandwidth} = \frac{100 \times 10^6}{10^3} = 10^4$$

$$\text{Minimum number of chips} = \lceil \log_2(10^4) \rceil = 13 \text{ chips}$$

Example 11.4

A communication system transmits at 120 kbps and uses 32-FSK (Frequency Shift Keying). A hop rate of 2000 hops per second is used over an available spectrum of 10 MHz. Assuming a negligible synthesizer switching time between hops, calculate (a) data symbol transmitted per hop, and (b) the number of nonoverlapping hop frequencies.

Solution

For 32-FSK, we have $32 = 2^b$ i.e., $b = 5$ bits per symbol

$$\text{Symbol rate} = \frac{120 \text{ kbps}}{5} = 24 \text{ kbps}$$

Since the symbol rate is higher than the hop rate, the system is a slow FHSS system.

$$\text{Number of symbols per hop} = \frac{24,000}{2,000} = 12$$

Minimum bandwidth B_w of an M-ary FSK $\sim MR_s$:

$$B_w = 12 \times 24 = 288 \text{ kHz}$$

Number of nonoverlapping hop frequencies:

$$n_{FH} = \frac{10 \text{ MHz}}{288 \text{ kHz}} \approx 35$$

Example 11.5

Consider an FHSS system in which the input data rate is 200 bits per second. The modulation scheme of 32-ary FSK is used to generate the modulation symbol. The frequency hopping rate is 200 hops per second. Calculate: (a) minimum separation between frequency tones; (b) number of frequency tones produced by a frequency synthesizer; (c) processing gain; and (d) hopping bandwidth. Assume a frequency multiplication factor $K = 1$.

Solution

With the 32-FSK modulation scheme there are 5 chips per symbol:

$$\text{Symbol rate: } R_s = \frac{200}{5} = 40 \text{ symbols/sec}$$

The hop rate is higher than symbol rate, the system is a fast FHSS system.

$$\text{Symbol duration} = \frac{1s}{40} = 25 \text{ ms}$$

$$L = \frac{200}{40} = 5 \text{ hops/symbol}$$

$$\text{Chip duration} = \frac{25}{5} = 5 \text{ ms}$$

$$\text{Minimum separation between tones} = \frac{1}{5 \times 10^{-3}} = 200 \text{ Hz}$$

$$M = 2^5 = 32 \text{ frequency tones}$$

$$\text{Frequency hopping bandwidth} = KM\Delta fL = 1 \times 32 \times 200 \times 5 = 64 \text{ kHz}$$

$$G_p = MKL = 32 \times 1 \times 5 = 160$$

11.6 Operational Advantages of SS Modulation

The following are the operational advantages of SS modulation:

- **Low probability of intercept:** Low probability of intercept implies that a third party cannot easily eavesdrop on the conversation or has to utilize expensive means to accomplish this. A standard communications receiver selects a demodulation circuitry, such as an amplitude demodulator, a frequency demodulator, or a phase demodulator, depending on the modulation scheme used at the transmitter. In an SS system, the receiver demodulates the transmitted energy through some correlation process and effectively combines various components within a wider bandwidth. A single-channel receiver will thus detect a small portion of the transmitted signal which will be too weak for normal detection, and even if it could be magnified to a detectable level would be incomplete and thus not possible to understand.
- **Low probability of position fix:** Conventional radio transmitters are easily pinpointed by simple and inexpensive direction finders. The spectral-spreading concepts make this task much more demanding as greater processing power and integration time will be required within each resolution.

- **Low probability of signal exploitation:** This refers to the possibilities which exist within a communication scenario to exploit the communication link by some manipulation of the waveforms used to carry the message. These could be:
 - destruction of synchronization messages;
 - destruction or alteration of the message contents;
 - invisible or concealed addition of data bits.
- **High resistance to jamming and interference:** SS systems are inherently more robust against jamming and interference than systems not using spreading techniques. This property of SS modulation is used in military systems where the main design objective is to develop a system which is able to deliver a message through a very hostile and impenetrable medium. SS modulation techniques provide an additional factor which is not seen in conventional systems. This is due to the fact that a code is used in the spreading process and unless the jammer/interferer manages to get hold of this code the impact of the jamming/interference is reduced by a significant amount. There are two very important aspects of SS with respect to jamming and interference. The first relates to the actual protection provided by an SS code. The SS receiver provides a post-detection signal-to-noise and signal-to-interference improvement. This means that if the SS signal can be received with sufficient clarity and strength and without interference signals, intelligent jammers might redirect the same waveform with more power and cause problems for the detection process in the data link. This is a practical limitation since a limited number of spreading codes are used in an SS modem and thus the transmission tends to repeat the same codes a large number of times. The vulnerability lies in the fact that the code might be revealed if it sticks out clearly only once—for instance if it is received at a very short distance from the transmitter. This points at the very important requirement of SS systems, that the spectral-spreading scheme should be sufficiently agile and use frequent change of code structures. The next important aspect of SS modulation with respect to jamming and interference is that SS modulation provides a practical way of coping with frequency-band congestion.
- **High time resolution/reduction of multipath effects:** Multipath effects are one of the unavoidable effects in radio communication. Multipath implies that the signal reaching the receiver antenna has travelled by two or more paths. Because these routes inevitably are of different lengths, the time delays of signals that have come along the respective paths are different and the signal will fade in or out with small displacements of the transmitter and receiver (displacements of the order of half a wavelength). Methods exist whereby the signal is coded such that the signals reaching the receiver via different paths add up in phase at the receiver. Adaptive methods capitalizing on the wider bandwidth of SS waveforms make it possible to use radio communication under extremely severe multipath conditions. Example of

such are wireless local area networks used inside rooms or buildings where the propagation conditions are very poor.

- **Cryptographic capabilities:** The coding aspects of SS modulation have implications for possible security function in a communication link. The spreading codes can be chosen such that they serve the dual purpose of spreading the frequency spectrum of the transmitted signal and making it difficult to decipher the message. This requires that the code provide the required spectral signature (usually reasonably flat), has good anti-cryptographic property, and is sufficiently redundant. As these criteria place rather tough requirements on the coding strategy, a more common approach is to implement SS coding and scramble the code for cryptography independently and usually sequentially.

11.7 Coherent Binary Phase-Shift Keying DSSS

The simplest form of a DSSS communication system uses coherent binary phase-shift keying (BPSK) for both data modulation and spreading modulation. But the most common form uses BPSK for data modulation and quadrature phase-shift keying (QPSK) for spreading modulation.

We first consider the simplest case. The i th mobile station is assigned a spreading code signal $c_i(t)$ (a periodic spreading code sequence with chips of width T_c). Each mobile station has its own such code signal. Information bits are transmitted by superimposing the data bits onto the code signal. If the i th mobile station transmits the binary data waveform $x_i(t)[x_i(t) = \pm 1]$, it forms the binary sequence.

$$m_i(t) = x_i(t)c_i(t) \quad (11.19)$$

Equation 11.19 represents the modulo-2 addition of $c_i(t)$ and $x_i(t)$ as a multiplication because the binary 0 and 1 represent values of 1 and -1 into the modulator.

The transmitted signal from the i th mobile is

$$s_i(t) = x_i(t)c_i(t)\sqrt{2P}\cos(\omega_c t + \phi) \quad (11.20)$$

where:

- $x_i(t)$ = baseband signal for the i th mobile
- $c_i(t)$ = spreading code for the i th mobile
- ω_c = carrier frequency
- P = signal power
- ϕ = data phase modulation

If T_b is the bit period of $s_i(t)$, then T_b may correspond to either a full period for $c_i(t)$, or to a fraction of a period. If T_b is less than one code period, then the data bits are modulating the polarity of a portion of a code period. The code $c_i(t)$ serves as a subcarrier for the source data. Since each mobile station uses the entire

channel bandwidth and since Equation 11.19 has a code chip rate of $1/T_c$ chips per second, each BPSK carrier uses an RF bandwidth of

$$B_w = \frac{1}{T_c} \quad (11.21)$$

The available channel RF bandwidth determines the minimum chip width, and the code period determines its relation to the bit time. The number of code chips per bit is given by:

$$G_p = \frac{T_b}{T_c} = \frac{B_w}{R_b} \quad (11.22)$$

The ratio B_w/R_b is the CDMA processing gain, G_p , or simply, the spreading ratio of code modulation. This shows how much the RF bandwidth must be spread relative to the bit rate, R_b , to accommodate a given spreading code length. Each mobile station (MS) uses the same RF carrier frequency and RF bandwidth, but with its own spreading code $c_i(t)$.

The signal in Equation 11.20 is transmitted using a distortionless path with transmission. The signal is received together with some type of interference and/or Gaussian noise. Demodulation is performed in part by remodulating with the spreading code appropriately delayed as shown in Figure 11.8. This correlation of the received signal with the delayed spreading waveform is the despreading. This is a critical function in all spread spectrum systems. The signal component of the output of the despreading is:

$$x_i(t - \tau_d) \sqrt{2P} c_i(t - \tau_d) \times c_i(t - \hat{\tau}_d) \cos(\omega_c(t - \tau_d) + \phi) \quad (11.23)$$

where:

$\hat{\tau}_d$ = receiver's best estimate of the transmission delay

Since $c_i(t) = \pm 1$, the product $c_i(t - \tau_d) \times c_i(t - \hat{\tau}_d)$ will be unity if $\tau_d = \hat{\tau}_d$, that is, if the code at the receiver is synchronized with the spreading code at the transmitter. When correctly synchronized, the signal component of the output of the receiver despreading is equal to $\sqrt{2P} x_i(t - \tau_d) \cos(\omega_c(t - \tau_d) + \phi)$, which can be demodulated using a conventional coherent phase modulator.

The bit error probability, P_e , associated with the coherent BPSK spread spectrum signal is the same as with the BPSK signal and is given as:

$$P_e = \frac{1}{2} \operatorname{erfc} \left[\sqrt{\left(\frac{E_b}{N_0} \right)_o} \right] = Q \left[\sqrt{2 \left(\frac{E_b}{N_0} \right)_o} \right] = Q \left[\sqrt{2 \left[G_p \cdot \left(\frac{E_b}{N_0} \right)_i \right]} \right] \quad (11.24)$$

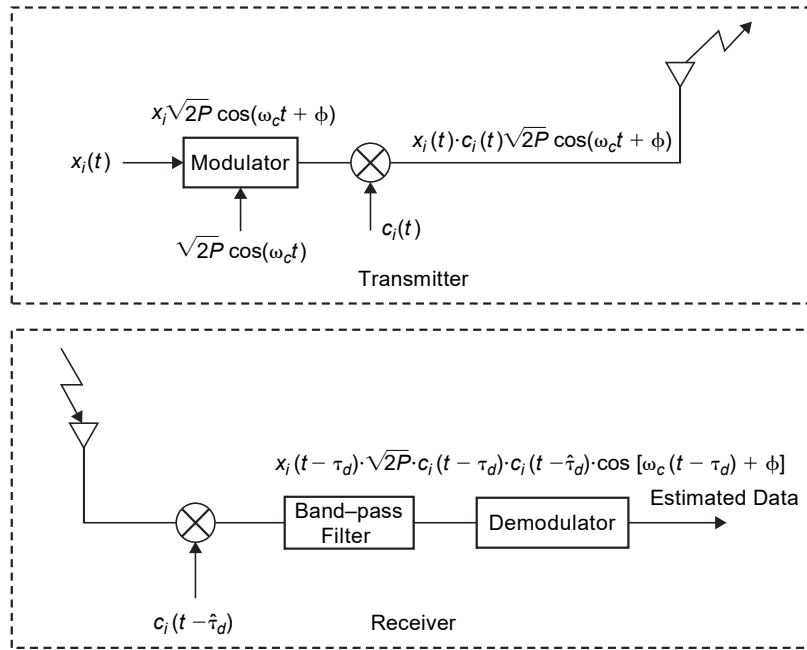


Figure 11.8 DSSS system with BPSK.

where:

$$Q(u) \approx \frac{e^{-u^2/2}}{(\sqrt{2\pi}u)}, \quad u \gg 1 \text{ is known as } Q \text{ function (see Appendix C)}$$

$$E_b = P/R_b = \text{energy per bit}$$

11.8 Quadrature Phase-Shift Keying DSSS

Sometimes it is advantageous to transmit simultaneously on two carriers which are in phase quadrature. The main reason for this is to save spectrum because, for the same total transmitted power, we can achieve the same bit error probability, P_e , using one-half the transmission bandwidth. The quadrature modulations are more difficult to detect in low probability of detection applications. Also, the quadrature modulations are less sensitive to some types of jamming. We refer to Figure 11.9 and write:

$$s(t) = \sqrt{P}c_I(t) \cos[\omega t + \phi] + \sqrt{P}c_Q(t) \sin[\omega t + \phi] \quad (11.25)$$

$$s(t) = a_I(t) + a_Q(t) \quad (11.26)$$

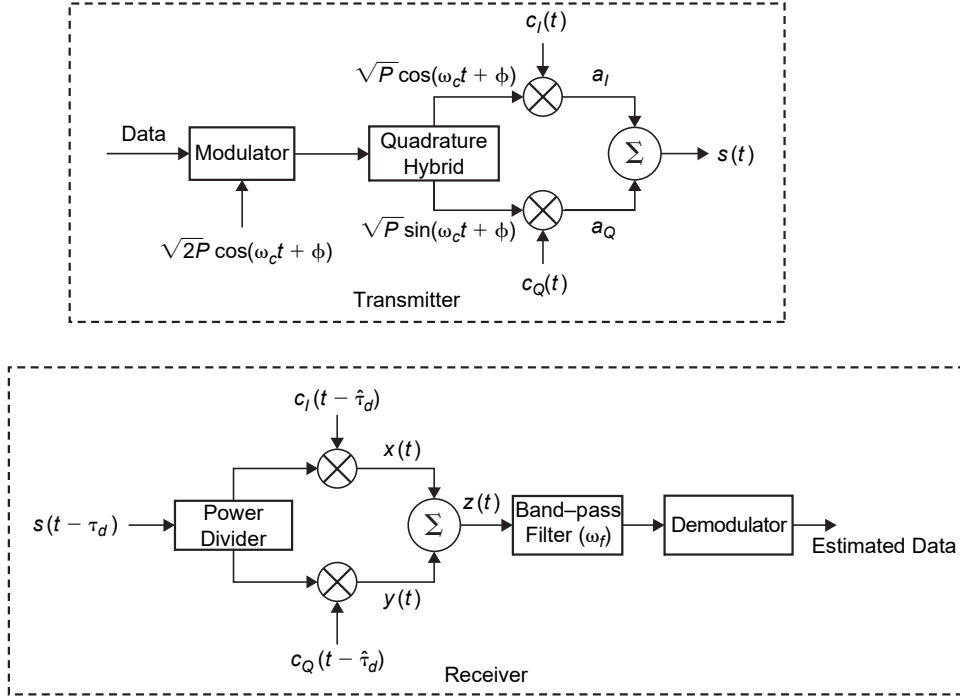


Figure 11.9 DSSS system with QPSK.

where:

$c_I(t)$ and $c_Q(t)$ are the in-phase and quadrature spreading codes and $a_I(t)$ and $a_Q(t)$ are orthogonal. This condition is satisfied in the present case since $c_I(t)$ and $c_Q(t)$ are independent code waveforms.

The receiver for the transmitted signal is shown in Figure 11.9. The bandpass filter is centered at frequency ω_f and has a bandwidth sufficiently wide to pass the data-modulated carrier without distortion.

$$x(t) = \sqrt{P/2}c_I(t - \tau_d)c_I(t - \hat{\tau}_d)\cos[\omega_c t + \phi] + (\sqrt{P/2})c_Q(t - \tau_d)c_I(t - \hat{\tau}_d)\sin[\omega_c t + \phi] \quad (11.27)$$

$$y(t) = \sqrt{P/2}c_I(t - \tau_d)c_Q(t - \hat{\tau}_d)\sin(\omega_c t + \phi) + (\sqrt{P/2})c_Q(t - \tau_d)c_Q(t - \hat{\tau}_d)\cos[\omega_c t + \phi] \quad (11.28)$$

If the receiver-generated replicas of spreading codes are correctly phased then

$$c_I(t - \tau_d)c_I(t - \hat{\tau}_d) = c_Q(t - \tau_d)c_Q(t - \hat{\tau}_d) = 1 \quad (11.29)$$

$$z(t) = x(t) + y(t) = \sqrt{2P}\cos[\omega_c t + \phi] \quad (11.30)$$

The signal $z(t)$ is the input to a conventional phase demodulator where data is recovered.

When the spreading codes are staggered one-half chip interval with respect to each other, the QPSK is called offset-QPSK (OQPSK). In OQPSK, the phase changes every one-half chip interval, but it does not change more than $\pm 90^\circ$. This limited phase change improves the uniformity of the signal envelope compared to BPSK and QPSK, since zero-crossings of the carrier envelope are avoided. Neither QPSK nor OQPSK modulation can be removed with a single stage of square-law detection. Two such detectors and the associated loss of signal-to-noise ratio are required. QPSK and OQPSK offer some low probability of detection advantages over the BPSK method.

11.9 Bit Scrambling

Referring to Table 11.4, we consider the following activities at a given transmitter location (see Figure 11.10):

1. An arbitrary data sequence $s_i(t)$ is generated by a digital source.
2. An arbitrary code sequence $c_i(t)$ is produced by a DS generator.
3. Two sequences are modulo-2 added and transmitted to a distant receiver.
4. At the distant location, the resulting sequence (assuming no propagation delay) is picked up by the receiver (see Figure 11.10).

Table 11.4 Operations with modulo-2 addition.

Transmitter	1	$s_i(t)$	1	1	0	1	0	0	1	1	1	1
	2	$c_i(t)$	1	0	0	1	1	1	0	1	0	0
	3	$s_i(t) \oplus c_i(t)$	0	1	0	0	1	1	1	0	1	1
Receiver	4	$s_i(t) \oplus c_i(t)$	0	1	0	0	1	1	1	0	1	1
	5	$c_i(t)$	1	0	0	1	1	1	0	1	0	0
	6	$s_i(t) \oplus c_i(t) \oplus c_i(t) = s_i(t)$	1	1	0	1	0	0	1	1	1	1

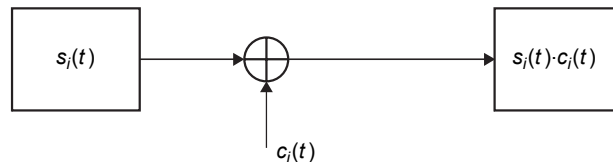


Figure 11.10 Mobile transmitter.

5. The code $c_i(t)$ used at the transmitter is also available at the receiver.
6. The original data sequence is recovered by modulo-2 adding the received sequence with the locally available code $c_i(t)$.

Next, referring to Table 11.5, we consider the following set of activities at the given transmitter location.

1. An arbitrary data sequence $s_i(t)$ is generated by a digital source. In this case, we use +1s and -1s to represent 0s and 1s.
2. An arbitrary code sequence $c_i(t)$ is produced by a DS generator.
3. We multiply $s_i(t)$ and $c_i(t)$. The output of the multiplier is transmitted to a distant receiver.
4. At the distant location, the resulting sequence (again assuming no propagation delay) is picked up by the receiver (Figure 11.11).
5. The code $c_i(t)$ used at the transmitting location is also available at the receiver.
6. The original data sequence is recovered by multiplying the received sequence by the locally available code, $c_i(t)$.

From Tables 11.4 and 11.5 we conclude that modulo-2 addition using 1s and 0s binary data is equivalent to multiplication using -1 and 1 binary data as long as we remain consistent in mapping 0s to +1s and 1s to -1s as shown in Table 11.5. (For circuit implementation, modulo-2 addition is preferred since exclusive OR

Table 11.5 Operations without modulo-2 addition.

Transmitter	1	$s_i(t)$	-1	-1	1	-1	1	1	-1	-1	-1	-1
	2	$c_i(t)$	-1	1	1	-1	-1	-1	1	-1	1	1
	3	$s_i(t) \cdot c_i(t)$	1	-1	1	1	-1	-1	-1	1	-1	-1
Receiver	4	$s_i(t) \cdot c_i(t)$	1	-1	1	1	-1	-1	-1	1	-1	-1
	5	$c_i(t)$	-1	1	1	-1	-1	-1	1	-1	1	1
	6	$s_i(t) \cdot c_i(t) \cdot c_i(t) = s_i(t)$	-1	-1	1	-1	1	1	-1	-1	-1	-1

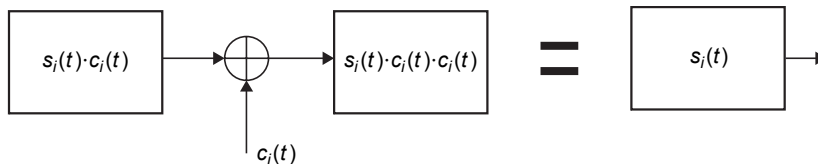


Figure 11.11 Mobile transmitter.

gates are cheaper than multiplication circuits. However, for modeling purposes, the multiplication method is usually easier to formulate and understand than the modulo-2 approach.)

We notice that for the output of the receiver to be identical to the original data, the following relationship must be satisfied:

$$s_i(t)c_i(t)c_i(t) = s_i(t) \quad (11.31)$$

In other words, $c_i(t)c_i(t)$ must be equal to 1. Note that $c_i(t)$ is a binary sequence made up of 1s and -1 s, therefore

$$\text{If } c_i(t) = 1, c_i(t)c_i(t) = 1 \quad (11.32)$$

$$\text{If } c_i(t) = -1, c_i(t)c_i(t) = 1 \quad (11.33)$$

In these discussions we assume that there is no propagation delay and no other processing delay incurred between the transmitter and receiver input. Thus the code copy used at the receiver is perfectly lined up with the initial code used at the transmitter. The two codes are said to be in phase or in synchronization. In practice, however, propagation delay and other processing delays (τ_i) occur between the transmitter and the receiver. Therefore, the receiver may be time-shifted relative to the initial code at the transmitter. The two codes are no longer in synchronization. As a result, the output of the receiver will no longer be identical to the original data, $s_i(t)$.

In order to recover the original data, $s_i(t)$, we must “tune” the receiver code sequence to that of the incoming code from the transmitter. In other words, we must time-shift the receiver code in order to line it up with the incoming code. It should be noted that by synchronizing or “tuning” the receiver code to the phase of the incoming code $c_i(t - \tau_i)$, the original data (shifted by propagation delay) can now be recovered at the output of the receiver.

In these discussions, the data sequence and code sequence are assumed to have the same length (one code bit for each data bit) and are used for encrypting the data bits. The process is referred to as *bit scrambling*.

11.10 Requirements of Spreading Codes

To spread the data sequence, the code sequence must be much faster than the data sequence, and exhibit some random properties.

By multiplying the data sequence with the faster code sequence, the resulting product yields a sequence with more transitions than the original data. Using

suitable random-like codes, the resulting sequence will have the same rate as the code sequence. It is desirable to use a set of orthogonal codes (see Appendix D) to provide good isolation between users. However, in practice, the codes used are not perfectly orthogonal, but they exhibit good isolation characteristics, i.e., they have low cross-correlation.

11.11 Multipath Path Signal Propagation and Rake Receiver

In the absence of a direct line-of-sight signal from the base station (BS) to the mobile station (or the received signal at the mobile station from the base station), the signal is made up of the sum of many signals, each travelling over a separate path. Since these path lengths are not the same, the information carried on the radio link experiences a spread in delay as it travels between the base station and the mobile. In addition, to delay spread, the same multipath environment causes severe local variations in signal strength as these multipath signals are added constructively and destructively at the receiving antenna. This effect is called *Rayleigh fading* (see Chapter 3). The movement of a mobile causes each received signal to be shifted in frequency as a function of the relative direction and speed of the mobile. This effect is called *Doppler shift* (see Chapter 3).

Multipath is treated as causing delayed versions of the signal to add to the system noise when the differential delay exceeds the chip time, T_c . Substantial performance improvement can occur by detecting each additional path separately, thereby enabling the signals to be combined coherently. A receiver can be implemented to resolve each individual path such that the paths can be combined to produce an overall gain. This type of receiver is known as a *Rake receiver*.

In the Rake receiver for user # 1 (refer to Example 11.2) baseband demodulated signal $Z(t)$ is the sum of N signals which arrives on N different paths (see Figure 11.12). We consider path 2, the multiplication of $Z(t)$ by $c_1(t - \Delta_2)$. The integration starts at time Δ_2 and ends at T_b , to yield the peak response for the path 2. (The output of the integrator is the value of the correlation function of $c_1(t)$ for a particular delay. For path 2, this delay is zero, whereas for the other paths the delay exceeds the time duration of a chip.) The contributions from other paths average out to 0, since the differential delays exceed the chip duration, T_c . The response from each path is summed to produce the stronger signal. We illustrate this concept with Example 11.6.

Example 11.6

We consider the downlink in Example 11.2 where the demodulated signal at the mobile is $z(t) = m_1(t) + m_2(t) + m_3(t)$ (see Figure 11.13(a)) for mobile 1, 2, and 3. We assume two equal-strength paths and write the demodulated signal as $Z(t) = z(t) + z(t - 2T_c)$ (see Figure 11.13(c)). The differential delay between these two paths is taken as $2T_c$ for simplicity. We will show how mobile 1 will